Energy Subsidy Reform and Manufacturing Firm Performance: The Roles of Technology and Capital Intensity

An Econometric Reassessment of Iran's 2010 Reform Using DiD and DDD

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Research Context

Iran's 2010 Energy Subsidy Reform (SRCT)



The Shock

In December 2010, Iran removed decades-old energy subsidies overnight — a policy shift often described as the largest economic surgery in the country's history.



Scale of Price Increases

Energy prices jumped sharply within months: kerosene rose about 6x, natural gas about 4x, while gasoline and electricity nearly doubled.



Universal Reform

The reform affected virtually all sectors at the same time, leaving no truly untreated group and creating a major identification challenge for empirical analysis.

Prices in real terms; energy carriers converted using Ministry of Energy guidelines.

Why Manufacturing Firms Matter

A pillar of the Iranian economy, exposed directly to the reform

17%

of overall GDP
(24% excl. oil revenues)

31%

of national
employment

~80%

of manufacturing value-added from highly / moderately energy-intensive industries

- Manufacturing value-added grew from ~400 to ~1,400 trillion Iranian Rial between 1994 and 2010 — a sustained upward trend interrupted sharply by the reform.
- Because the shock hit every firm at once, isolating its causal effect requires a carefully constructed counterfactual — the central empirical challenge this literature addresses.

Research Question

How did the 2010 energy subsidy reform affect the output, value-added, and operating surplus of Iranian manufacturing firms, and to what extent were these effects moderated by technology and pre-reform capital intensity?

Data Source

- Statistical Center of Iran (SCI) annual survey, firms with 10+ workers
- Firm-group level (4-digit ISIC × province), 2009–2013
- Unbalanced micro-panel, 5,636 group-year observations

Three Outcome Indicators

Output

Produced goods + manufacturing services

Value-added

Output minus input

Operating-surplus

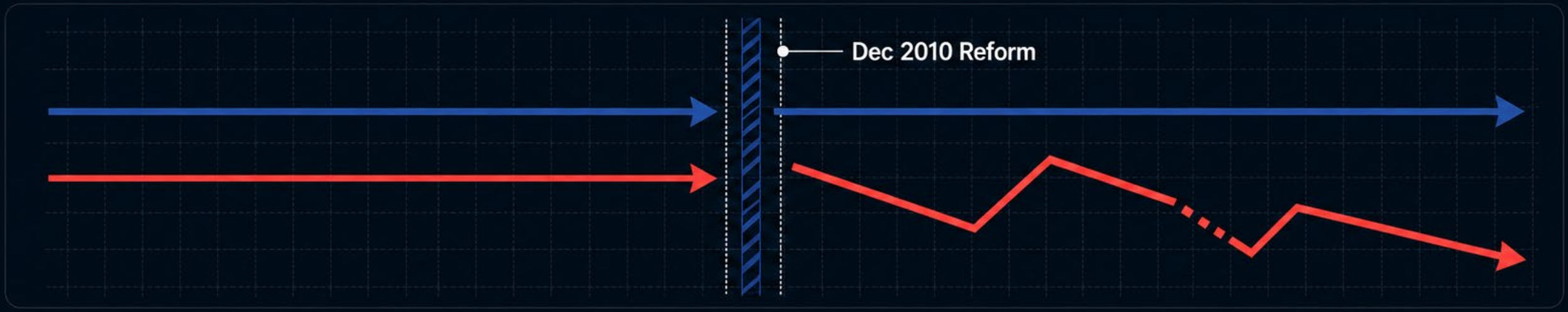
Value-added minus compensation (profit proxy)

All indicators in logarithmic, real (2011=100) form.

Isolating the shock requires a precise, counterfactual-driven design.

Difference-in-Differences (DID) Architecture





Treatment Group (The Exposed)

Highly energy-sensitive industries facing the brunt of the aggressive price hikes.

Control Group (The Counterfactual)

Less energy-sensitive sectors (excluding machinery/vehicles) used to cleanly model what would have happened in the absence of the shock.

Methodology

$$Y_{it} = \alpha_i + X_{it}\theta + \delta Oil_t + \tau Trend_t + \varepsilon_{it}$$

Step 1: Before–After

$$Y_{it} = \alpha_i + \beta\, Post_t + \theta' X_{it} + \delta Oil_t + \tau\, Trend_t + \varepsilon_{it}$$

- Post = 1 for 2011–2013
- Within-unit descriptive change

Step 2: Difference-in-Differences

$$Y_{it} = \alpha_i + \beta(Post_t \times Treat_i) + \theta' X_{it} + \delta Oil_t + \tau\, Trend_t + \varepsilon_{it}$$

- Treat = energy-sensitive industries
- β = differential post-reform effect

Step 3: DDD Moderation

$$Y_{it} = \ldots + \eta(Post_t \times M_i) + \kappa(DID_{it} \times M_i) + \varepsilon_{it}$$

- M = technology or pre-reform capital intensity
- κ = difference in the DiD effect across M

Unit fixed effects • Firm controls • Oil-export proxy • Linear trend • Unit-clustered SE • No year fixed effects

Estimated losses deepen across industrial performance measures.



A statistically significant relative decline in output.

A larger and strongly significant decline in value-added.

The largest point estimate, but significant only at the 10% level.

Key Insight: Under the paper's no-year-FE specification, the reform is associated with relative declines in Output (-3.35%, p = 0.0430), Value-Added (-7.30%, p = 0.0098), and Operating Surplus (-9.12%, p = 0.0602). Output and value-added are significant at 5%; the operating-surplus result is marginal at 10%.

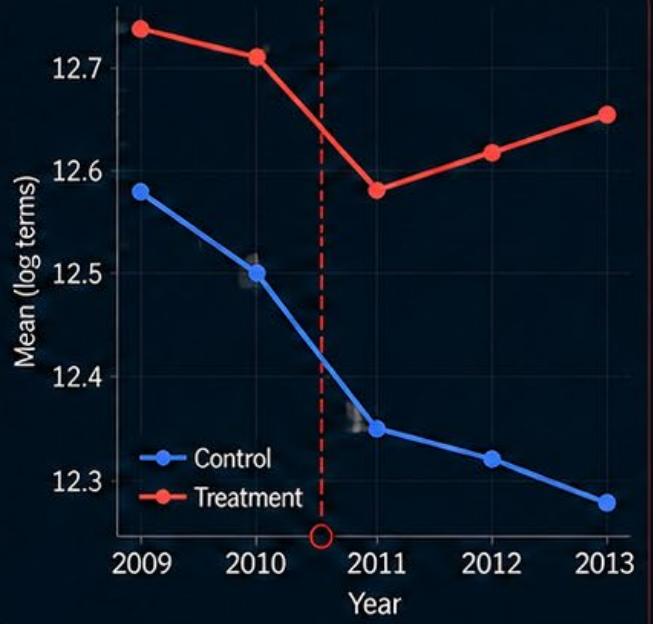
*** p < 0.01 ** p < 0.05 * p < 0.10

Raw data traces an immediate, stark divergence in firm survival.

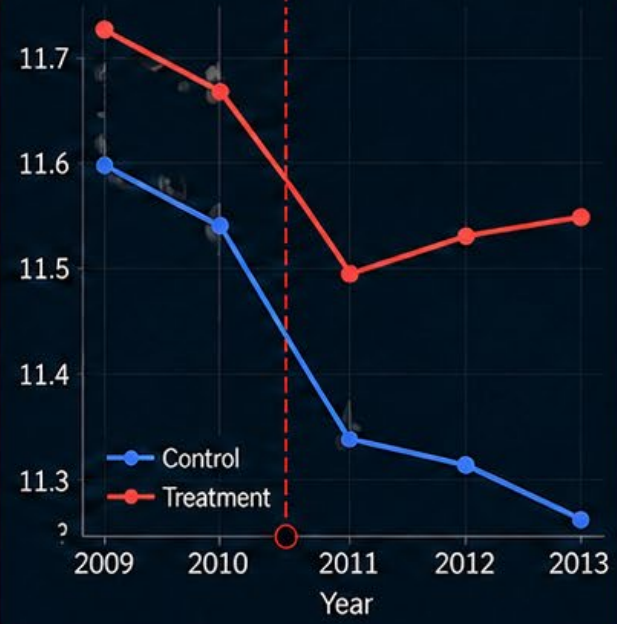
The parallel trajectory between energy-sensitive (red) and less-sensitive (blue) industries holds firmly through 2010.

By the first full year post-reform (2011), the treatment group experiences a sharp relative decline across all three performance metrics, visually confirming the causal footprint of the policy.

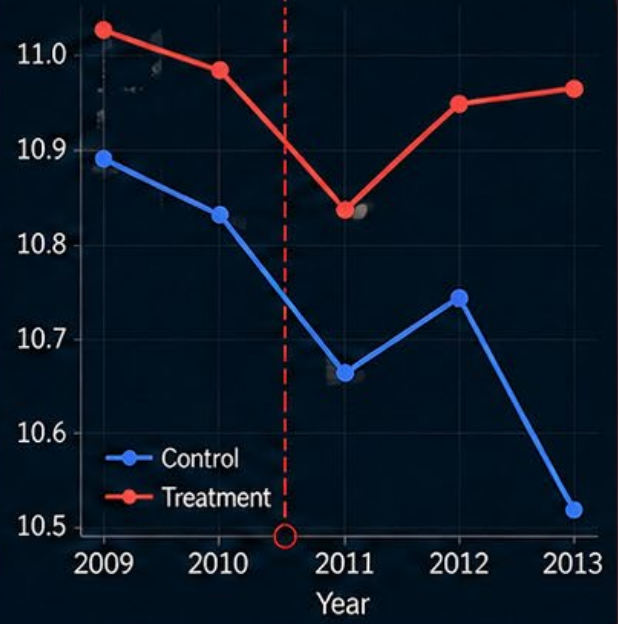
Output (log)



Value-added (log)



Operating-surplus (log)



Key Insight: Across output, value-added, and operating-surplus, the treatment group (energy-sensitive industries) experiences a sharp relative decline starting in 2011. All three baseline effects are negative and statistically significant at the 1% level.

Transmission Channels

How energy prices reach firm performance

Direct Channel

Higher direct energy costs (fuel, electricity) raise marginal costs immediately.

Indirect / Pass-Through

Energy-embodied intermediate inputs (metals, cement) become costlier — a ‘snowball’ along the production chain.

Demand Contraction

Higher prices reduce households' real income, shrinking aggregate demand for goods.

Notably, the indirect (pass-through) channel proves at least as important as the direct channel — moderately energy-intensive downstream industries are hit hardest.

Three Identification Challenges in the Base Design

What the baseline estimates do—and do not—identify

1. Contemporaneous Inputs as Controls

The base specification conditions on raw materials, energy, capital formation, wages, and firm count.

These inputs may themselves respond to the reform.

Therefore, the SRCT coefficient is a conditional effect holding observed input use fixed—not necessarily the total reform effect.

⚠ Potential post-treatment controls

2. Imbalanced Comparison & Inference

Control observations: 668 of 5,636 (≈12%).

Treatment exposure varies mainly at the 2-digit ISIC level (≈22 industries), while standard errors are clustered at the firm-group × province level (≈1,300 units).

Clustering below the assignment level may yield overly precise inference.

⚖ Assignment and clustering levels differ

3. Limited Adjustment for Common Shocks

The paper-aligned model intentionally includes no year fixed effects.

A linear trend and a national oil-export/ sanctions proxy absorb common time shocks only parametrically.

Year-specific or industry-specific shocks may remain in the estimated effect.

🕒 No year FE in the main specification

Interpretation: These issues do not invalidate the replication, but they narrow its causal interpretation.
The estimates should be presented as paper-aligned conditional effects and accompanied by sensitivity and robustness checks.

Our Design: Continuous Pre-Reform Exposure

A dose-based specification with stronger control for industry and regional shocks

1. Construct Pre-Reform Exposure

Direct exposure

2009–2010 mean energy-cost share of output

Sample mean: 2.0% (the paper reports 1.6%)

Indirect exposure

Non-energy intermediate-input share: materials + packaging + tools

Measured before treatment

2. Prepare the Exposure Measures

- Winsorize at the 1st and 99th percentiles
- Standardize to mean 0 and SD 1
- Use rank-based exposure as a robustness check

Estimation sample:

1,213 firm-groups

5,426 observations

123 groups dropped for missing pre-reform data

Pre-treatment dose is fixed over time

3. Estimate the Continuous Model

Equation (2c)

$$Y_{it} = \beta(Post_t \times DirectDose_i) + \gamma_i + \delta(ISIC2 \times year) + \delta(province \times year) + \varepsilon_{it}$$

- β measures how the post-reform change varies with one additional unit of pre-reform exposure.
- Replace $DirectDose$ with $IndirectDose$ for the pass-through specification.

Firm-group FE + industry-year FE + province-year FE

Identification Gain The binary treatment design cannot separately identify treatment with ISIC2 × year fixed effects because treatment is defined at the 2-digit industry level. Continuous exposure varies within industries, so these fixed effects can be included while β remains identified.

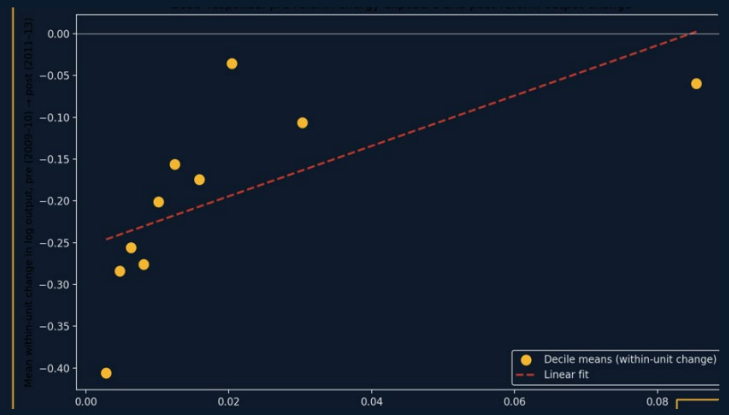
Restricted-sample binary DiD closely matches the full-sample estimate for output (−0.032 vs. −0.034), indicating that sample restriction is not driving the baseline result.

Results: Exposure Predicts a Less Negative Output Response

Dose-response evidence and event-study diagnostics

Dose–Response Gradient

Pre-reform energy share vs. post-reform output change



Specification 2c: Post × Dose

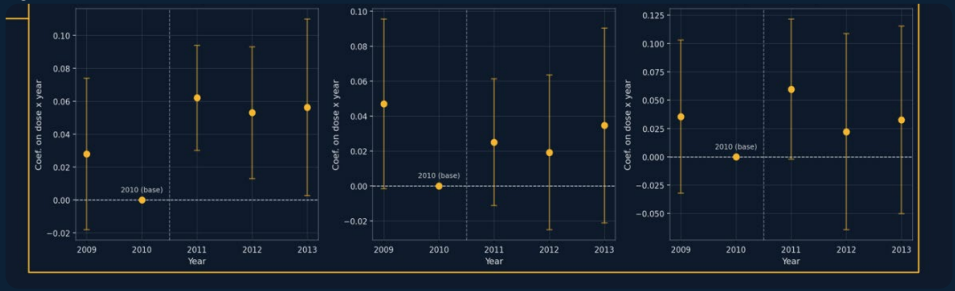
Output	**+0.051**	SE 0.021 • p = 0.015 *Statistically significant*
Value-added	**+0.012**	SE 0.022 • p = 0.58 *Not statistically significant*
Operating surplus	**+0.050**	SE 0.038 • p = 0.19 *Not statistically significant*

A one-SD increase in pre-reform energy exposure is associated with a less negative relative output response—not an absolute increase in output.

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Event-Study Coefficients

2010 is the omitted base year



Pre-Trend Diagnostics

OUTPUT — 2009 **+0.028 (SE 0.023; p = 0.23)**

Small and statistically insignificant.

VALUE-ADDED — 2009 **+0.047 (p = 0.058)**

Borderline evidence; retained as an explicit caveat.

Interpretation Output declines across all exposure groups, but the decline is markedly smaller at higher pre-reform exposure. Once industry-year and province-year shocks are absorbed, own energy intensity does not predict larger losses—consistent with an important indirect/pass-through channel.

Mechanisms: Direct Costs, Pass-Through, and Firm Adjustment

Joint exposure estimates distinguish supported mechanisms from remaining proxies

Direct vs. Indirect Exposure: Joint Model

Both pre-reform doses enter jointly with ISIC2 × year and province × year fixed effects

OUTCOME	DIRECT DOSE	INDIRECT DOSE	WALD: EQUALITY
Output	**+0.057** SE 0.026; p = 0.03	**+0.014** SE 0.032; p = 0.66	**p = 0.111** *Cannot reject equality*
Value-added	**+0.086** SE 0.027	**+0.165** SE 0.030	**p = 0.002** *Indirect dose is stronger*

What the Joint Test Supports

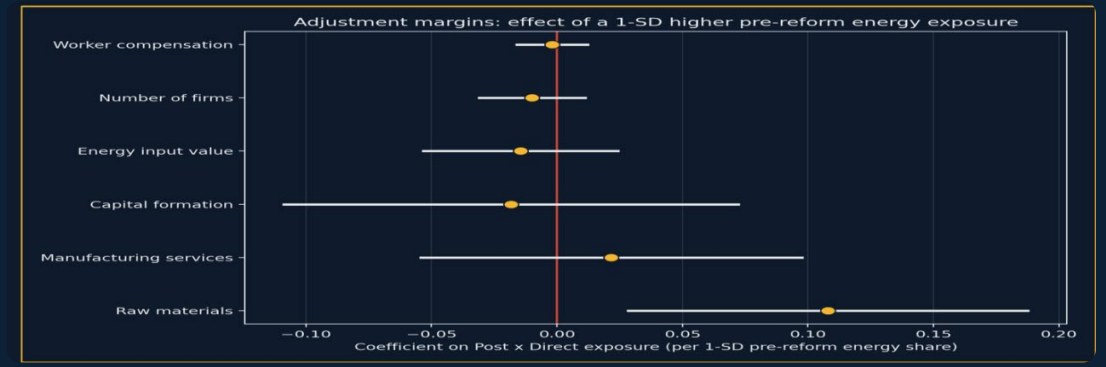
OUTPUT Direct exposure is significant, but the direct–indirect difference is not.

VALUE-ADDED Indirect exposure is statistically stronger (Wald p = 0.002), with a mechanical accounting caveat.

DOSE CORRELATION Direct–Indirect = −0.46; no obvious multicollinearity red flag.

How Firms Adjusted

Effect of a one-SD increase in pre-reform direct exposure



Adjustment & Absorption

RAW MATERIALS **+0.108 • p = 0.009**

The only statistically significant operating-margin response.

OTHER MARGINS

Energy −0.014; capital formation −0.018; wages −0.002; firm count −0.010; manufacturing services +0.022 — all insignificant.

OPERATING-SURPLUS ABSORPTION **Post × Margin = −0.402; p < 0.001**

Measurement caveat IndirectDose is a proxy. A stronger measure should use an Iranian input–output table to weight suppliers by energy intensity; the present results are suggestive, not a definitive decomposition of the transmission channel.

From the Base Paper to Our Extension

Four contributions that strengthen identification and mechanism testing

1



1. Total vs. Conditional Effects

The base paper conditions on contemporaneous inputs that may respond to the reform.

Our extension distinguishes the paper's input-conditional estimate from a broader total-effect benchmark.

Clarifies the estimand

2



2. Continuous Pre-Reform Exposure

Firm-group exposure measured in 2009–2010 replaces the highly imbalanced binary comparison.

Within-industry variation allows ISIC2 × year and province × year fixed effects.

Stronger control for common shocks

3



3. Formal Test of Transmission Channels

Direct and indirect exposure enter jointly.

A **Wald test** evaluates coefficient equality, whereas the base paper compares subgroup estimates without formally testing their difference.

Tests the paper's channel claim

4



4. New Evidence on Adjustment Margins

We estimate responses in energy use, capital formation, wages, raw materials, manufacturing services, and firm counts.

This reveals how firms adjusted—not only whether performance changed.

Extends outcomes beyond performance



Extension, Not Rejection

Our study builds directly on the base paper: it **clarifies the estimand, strengthens identification** through continuous pre-treatment exposure, **formally tests mechanisms,** and **adds evidence on adjustment margins.**

⚠ Indirect exposure remains a proxy; a supplier-weighted Iranian input–output measure is the natural next step.

Thank you for your attention

Group 16

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